

HOLMON, SwedenWMO: **022870**Lat: **63.792N**Lon: **20.861E**Elev: **49**StdP: **14.67**Time Zone: **1.00 (EUC)**Period: **05-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-2.0	3.3	-5.5	4.1	-1.5	-0.3	5.4	3.5	33.3	25.1	30.9	24.3	7.4	40	0.744

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	73.9	65.8	70.2	63.6	67.6	61.7	67.3	72.4	64.7	69.1	62.8	66.6	9.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.1	93.1	70.6	62.8	85.7	67.6	60.9	80.1	65.1	31.8	72.2	29.8	69.1	28.4	66.4	73.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.7	25.8	22.7	DB	-4.3	78.4	8.3	4.7	-10.3	81.8	-15.2	84.5	-19.8	87.1	-25.9	90.5	(4)
(5)			WB	-4.7	68.3	8.2	2.8	-10.7	70.3	-15.5	72.0	-20.1	73.5	-26.1	75.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	40.8	23.6	21.8	27.1	35.5	45.0	53.7	61.6	60.1	52.4	42.3	35.1	29.9
	DBStd	14.82	8.48	10.77	7.85	4.84	5.32	4.52	4.87	4.42	4.64	5.19	5.61	7.10
	HDD50	4286	819	790	710	436	172	16	0	0	30	242	448	623
	HDD65	8872	1284	1210	1175	886	619	338	130	162	378	704	898	1088
	CDD50	924	0	0	0	0	18	128	359	314	102	3	0	0
	CDD65	35	0	0	0	0	0	0	25	10	0	0	0	0
	CDH74	111	0	0	0	0	1	0	82	28	0	0	0	0
	CDH80	10	0	0	0	0	0	0	8	2	0	0	0	0
Wind	WSAvg	11.0	11.8	11.2	10.7	10.5	10.5	10.4	9.6	10.0	10.4	12.2	12.8	12.4
Precipitation	PrecAvg	24.5	2.0	1.6	1.4	1.3	1.6	1.9	2.4	2.8	2.2	2.4	2.6	2.2
	PrecMax	36.1	4.6	3.3	2.5	3.7	3.6	3.7	5.2	5.2	4.4	5.0	5.2	4.3
	PrecMin	16.2	0.4	0.1	0.3	0.3	0.1	0.6	0.4	0.5	0.4	0.2	0.7	0.5
	PrecStd	4.4	0.9	0.9	0.6	0.9	0.8	0.9	1.3	1.3	1.1	1.2	1.3	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.6	40.1	43.5	51.6	65.3	69.8	80.9	77.6	64.8	54.5	46.8	42.2
		MCWB	37.2	36.7	37.8	43.6	53.8	58.0	68.2	67.6	58.9	50.9	45.1	40.2
	2%	DB	37.0	37.3	40.1	48.0	59.3	65.9	75.9	71.6	62.3	52.4	45.7	40.6
		MCWB	35.7	35.0	35.7	41.5	50.5	57.0	67.4	65.0	57.3	50.2	44.3	39.2
	5%	DB	35.6	35.4	38.0	44.8	56.0	63.4	72.3	69.0	60.5	50.9	44.2	39.5
		MCWB	34.3	34.1	34.7	39.7	48.7	55.4	65.2	63.1	56.5	48.9	43.0	38.1
	10%	DB	34.1	33.9	36.1	42.3	53.0	61.3	69.3	66.7	58.8	49.4	42.6	38.2
		MCWB	33.1	33.0	33.9	38.1	47.1	54.3	63.4	61.4	55.5	47.5	41.4	37.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.8	37.1	38.5	45.0	54.8	60.3	70.8	70.3	60.7	52.5	45.9	41.1
		MCDB	38.2	39.4	42.3	50.2	62.5	66.2	77.0	75.7	63.2	53.3	46.4	41.6
	2%	WB	36.2	35.5	36.5	42.2	51.5	58.2	68.5	66.0	58.8	50.9	44.6	39.6
		MCDB	36.8	36.7	38.9	47.0	57.8	63.7	74.0	69.9	60.9	52.0	45.5	40.3
	5%	WB	34.6	34.3	35.3	40.3	49.5	56.7	66.3	64.0	57.4	49.2	43.1	38.4
		MCDB	35.4	35.3	37.1	44.2	55.0	61.9	71.3	67.6	59.4	50.5	44.0	39.3
	10%	WB	33.3	33.2	34.2	38.5	47.6	55.4	64.1	62.4	56.2	47.6	41.7	37.1
		MCDB	34.1	33.9	35.8	41.6	52.2	60.1	68.7	65.9	58.2	49.0	42.4	38.1
Mean Daily Temperature Range	5% DB	MDBR	7.7	9.3	10.6	10.1	11.0	11.3	11.1	9.7	8.8	6.9	5.8	6.4
		MCDBR	6.7	7.1	10.6	14.0	16.0	14.3	14.7	12.3	10.7	7.5	5.7	6.3
		MCWBR	6.5	6.1	7.6	8.7	8.8	7.7	7.2	7.0	6.7	6.0	5.5	6.0
	5% WB	MCDBR	6.6	6.4	8.8	13.3	14.9	12.7	12.8	10.8	8.9	6.5	5.7	6.3
		MCWBR	6.4	5.7	6.8	8.7	8.8	7.5	7.2	6.6	6.1	5.9	5.6	6.2
Clear-Sky Solar Irradiance	taub	0.226	0.287	0.311	0.348	0.350	0.350	0.368	0.357	0.329	0.312	0.362	0.351	
	taud	2.023	2.161	2.232	2.298	2.451	2.493	2.465	2.492	2.548	2.389	2.628	2.587	
	Ebn at Noon	114	198	246	262	274	276	266	259	245	193	106	57	
	Edh at Noon	9	19	27	32	30	30	30	27	21	16	8	5	
All-Sky Solar Radiation	RadAvg	41	221	698	1212	1631	1839	1682	1273	742	280	66	18	
	RadStd	4	43	79	115	154	145	153	114	77	53	12	2	